

The Power of Infinity: Cantor, Cohen, and Infinite Sets

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1 Introduction

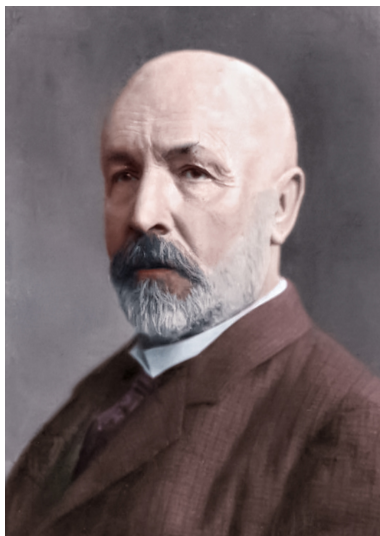


Figure 1: Portrait of Georg Cantor

This article examines some of the interesting and bizarre properties of infinite sets. Thanks to the work of Georg Cantor, a German mathematician of the late 19th and early 20th centuries, set theory began to develop rapidly. His ideas about infinite sets were so shocking and counterintuitive that they were often met with heavy skepticism from the mathematical world. People were outraged at this new and unconventional way of thinking, and even Henri Poincaré called this unusual way to think about sets and to conduct set theory proofs a “disease” in mathematics. In modern math, set theory is considered to be a very important branch, forming the foundation for many mathematical concepts and having many implications outside the world of pure math.

2 Infinite Sets and Bijections

Simply speaking, a set is a collection of elements or objects. Most of Cantor’s interesting work was focused on the particular infinite sets that allow us to group numbers: natural numbers, integers, rational numbers, irrational numbers, real numbers, transcendental numbers, and so on. One of Cantor’s methods for his exotic assertions would be to develop a **bijection**, or a one-to-one correspondence between two sets, to assert their equivalence. In other words, a bijection creates a relationship between two sets, say X and Y , such that each element from set X is paired with exactly one element from set Y , and vice versa.

For infinite sets, this story is more complex, and Cantor later introduced the concept of a cardinal number—a way to distinguish and to describe the various sizes of infinite sets. Certainly, the size of a finite set is just a number that describes the number of elements in that set. The set $\{1, 3, 5, 7, 9\}$, for instance, is a finite set that has a size of 5. However, for the description of the sizes of various infinite sets, the cardinal numbers are denoted using the Hebrew symbol aleph ($\aleph$) followed by a subscript. In the proofs that follow, it is worth noting that cardinality is defined in terms of bijections. Two sets have the same cardinality if and only if there is a bijection, or a one-to-one correspondence, between them.

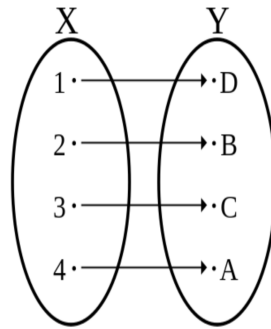


Figure 2: Example of a bijection

3 The Set of Natural Numbers and Integers

Suppose we consider the set of integers ($\mathbb{Z}$) and natural numbers ($\mathbb{N}$). A question arises: Is the set of integers larger than the set of natural numbers? Both of these sets go on forever, but it is clear that the set of natural numbers is a subset of the set of integers. After all, each natural number belongs to the set of integers, but not every integer belongs to the set of natural numbers, suggesting that there must be more integers than natural numbers. Shouldn't the set of integers be a "larger infinity"?

Of course, there is also an argument to the contrary. Suppose we have an infinite set of natural numbers, and we decide to add the negative of each element in the set to it to produce the set of integers. But will adding more elements to an already infinite set change that infinity quantity?

Which of these arguments is right? Although each makes important points, Cantor resolved this with a bijection argument. As one of the most fundamental assertions of infinite set theory, Cantor claimed that the cardinality of the natural number set ($\mathbb{N}$) is equal to that of the set of integers ($\mathbb{Z}$).

To prove his assertion, Cantor established a bijection between two sets, as follows. Arranging all of the integers in a list as 0, 1, -1, 2, -2, 3, -3 ... allows us to count them by creating a function between them and natural numbers. This can be described with the function that has an input of a natural number and an output of a unique integer. In fact, the natural number input is used to produce the position of each integer on the list. For example, the integer 0 is in position 1, 1 in position 2, and -1 in position 3, and so on. Therefore, we obtain:

$$\begin{aligned}
f(1) &= 0 \\
f(n) &= \frac{n}{2} \text{ for even values of } n \\
f(n) &= \frac{-(n-1)}{2} \text{ for odd values of } n
\end{aligned}$$

Let's check if this function works. For every natural number n we can find a corresponding integer in the list, and cover the entire set of integers. And for every integer we can find a corresponding natural number that is the position of that integer on the list. Thus, we have a bijection.

Note: Sets that are the same size as the set of natural numbers are considered the smallest infinite sets and are denoted with the cardinality of $\aleph_0$. These are also known as **countable sets**.

4 The Hilbert Hotel and Rational Numbers

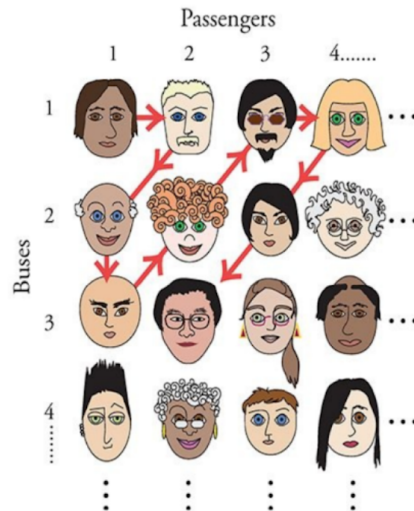
German mathematician David Hilbert, a great admirer of Cantor's work, once told a story of an interesting hotel known now as the Hilbert Hotel. This great hotel was always booked, and yet it always had a vacancy, for there were an infinite number of rooms in the Hilbert Hotel. And if a new guest arrived, all that the manager of the hotel had to do was to move the people from Room 1 to Room 2, Room 2 to Room 3, Room 3 to Room 4, and so on, thus freeing up the first room for the newcomers. Of course, if infinitely many newcomers arrive, the manager can always move the occupants of Room 1 to Room 2, Room 2 to Room 4, Room 3 to Room 6, and so on, thus freeing up all of the odd-numbered rooms at the Hilbert Hotel. Infinitely many odd-numbered rooms for infinitely many demanding newcomers. This manager must be quite pleased with himself.

Now the challenge comes: Infinitely many buses with infinitely many people, all clamoring noisily for a room, ride up to the Hilbert Hotel. Our good-natured manager (although growing less so by the minute) decides to perform the same trick as last time: to vacate infinitely many rooms, all odd, with his "doubling" idea. But will this suffice? We do, after all, have infinitely many rooms, and the number of passengers seems to be of the magnitude of infinity squared...

The trick here is this: If the manager were to start tackling this problem like the previous problem with one bus by trying to assist Bus 1, he would never get through all of the buses. For there are infinitely many passengers in Bus 1, and the rowdy people of Bus 2, for instance, would never see the warm, fluffy beds of the Hilbert Hotel.

The manager takes the problem in stride: He lines up the passengers in an infinite rectangular grid, with passengers and buses, and develops a foolproof algorithm for attending to them all. He begins with Passenger 1 on Bus 1 and gives her the first empty room. The second and third empty rooms go to Passenger 2 on Bus 1 and Passenger 1 on Bus 2, shown on the second diagonal from the corner of the diagram. The manager then tackles the third diagonal of passengers and assists Passenger 1 on Bus 3, Passenger 2 on Bus 2, and Passenger 3 on Bus 1. Notice, in the way that the passengers are processed diagonally, it is ensured that each particular person will be reached in some finite number of steps. If you provide some X number of steps, I can find the particular

person on this diagram that corresponds, and if you point out a person, I can identify a unique number of steps! And thus, the saying goes, “There’s always room at the Hilbert Hotel.”



Mathematically, this is simply a proof that a countable number of countable sets (like an infinity by infinity grid) will be countable, but its implications are massive, particularly with the cardinality of the set of rational numbers. Suppose we were to write out the set of rational numbers like this:

1/1	1/2	1/3	1/4	...
2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
⋮	⋮	⋮	⋮	⋮

Looks familiar? This table encompasses all rational numbers as we move through all of the possible numerators and denominators. Once again, we can move through our table diagonally.

1/1	1/2	1/3	1/4	...
2/1	2/2	2/3	2/4	...
3/1	3/2	3/3	3/4	...
4/1	4/2	4/3	4/4	...
⋮	⋮	⋮	⋮	⋮

Most importantly, we can reach any element in this table in a unique, finite number of steps; but what are those steps, really? The set of steps is equivalent to the set of natural numbers, since these steps are recorded that way: 1 step for 1/1, 2 steps for 2/1, 3 steps for 1/2, 4 steps for 1/3, and so on. This is another bijection! We can establish a one-to-one correspondence between the steps, or the set of natural numbers, and the set of rational numbers. And we know this must be a bijection because a number of steps determines a unique rational number, and a rational number determines a unique number of steps, thus proving the equivalent cardinalities of the sets $\mathbb{N}$ and $\mathbb{Q}$.

5 Real Numbers

How about real numbers $\mathbb{R}$? Could it be true that the number of real numbers and natural numbers is the same, just like the sets $\mathbb{Z}$ or $\mathbb{Q}$? The argument that proved the equivalence between integers and natural number sets was hard to accept because the natural number set is a subset of $\mathbb{Z}$. Can we use the same argument here, despite the fact that there are far more real numbers than integers (or are there)?

This proof by Cantor (one of his finest) shows that the number of real numbers in the interval $(0, 1)$ is uncountable and is done by contradiction.

Suppose we were to create the bijection between natural numbers and real numbers, such that each real number between 0 and 1 appears somewhere in this list.

For instance, our list could start like this:

1. .456789...
2. .987654...
3. .675675...
4. .123356...
- . . .
- . . .
- . . .

Cantor claimed he could produce a number that was not on the list.

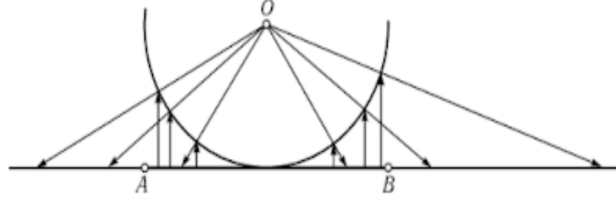
Suppose that we take the first digit from the first number, the second digit from the second number, etc. In other words, take numbers diagonally from the list. The decimal generated from the provided example would be .4853. The next step is to change each digit of this number to any other digit. For example, change the 4 to a 5, the 8 to a 9, the 5 to a 6, and the 3 to a 4. This yields a new number, .5964. This number is our winner. It cannot be the first number on our list, since it has a different first digit from the number there. It is also not the second number, since its second digit is different, etc. If instead we were to make an infinitely long list, implement Cantor's 'diagonal argument,' and change up each digit, we would be making a number that differs from the n th number in the n th decimal place. Ergo, it doesn't appear anywhere on the list, and there is no way that our list can cover all real numbers.

This was explosive in the math world: Cantor's clear proof and identification that the infinite size of the set of real numbers is greater than that of natural numbers. It was then assigned the cardinality $\aleph_1$, greater than $\aleph_0$. Sets with cardinality $\aleph_1$ are also referred to as **uncountable sets**, or continuums.

5.1 Real Numbers in an Interval Versus Real Numbers on the Entire Number Line ($\mathbb{R}$)

We have established that there are more real numbers on an interval $(0, 1)$ than there are natural numbers. But can we compare two sets of real numbers? Specifically, the number of real numbers in a fixed interval, like $(0, 1)$, versus the number of real numbers on the entire number line (two continuums)? Suppose we took the interval $(0, 1)$, picked it up like a string and curved it into a semicircle, tangent to the real number line at its midpoint. We could then place a laser

or a flashlight in the center of the semicircle and shine rays of light at it. The laser will hit the string at some point, continue past it, and then hit the number line. And every point on the string will have a corresponding point on the number line. This can be done for any point on the string, establishing the needed one-to-one correspondence, and proving that the cardinality of any two infinite sets of real numbers is equal to $\aleph_1$.



6 Connection Between $\aleph_0$ and $\aleph_1$

So far, we have focused on the countable set ($\aleph_0$) of $\mathbb{N}$, and the uncountable set ($\aleph_1$), or the continuum of real numbers, as very separate entities. But how can we tie them together?

For this, let us return briefly to finite sets and introduce the concept of the **power set**. The power set, $\mathcal{P}(S)$, of a set S is defined as the set of all subsets of S . For instance, take a set $S(a, b, c)$. The power set, $\mathcal{P}(S)$, is $[(), (a), (b), (c), (a, b), (a, c), (b, c), (a, b, c)]$. Note that $\mathcal{P}(S)$ includes the empty set as well. In this case, there were 3 elements in set S and 8 elements in $\mathcal{P}(S)$. Is there some relationship?

It is easy to see that for set $\mathcal{P}(S)$ above, the number of elements is

$$\binom{3}{0} = \binom{3}{1} + \binom{3}{2} + \binom{3}{3},$$

since we are choosing all possible numbers of elements from S , which has 3 elements. To generalize: If there is a set with n elements, its power set can be described as containing

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}$$

elements. Using the theorem about the sum of the n th row of Pascal's triangle, we can conclude that the power set has 2^n elements. Also of note, the power set must always have more elements than the original set, since $2^n > n$ for all positive n . Therefore, it follows that the power set of the set of natural numbers, $\mathcal{P}(\mathbb{N})$, has $2^{\aleph_0}$ elements, and this power set has more than $\aleph_0$ elements.

Now consider the set of real numbers on the interval $(0, 1)$, but this time, consider it in terms of binary, with each decimal made up entirely of 0s and 1s after the decimal point. The number of digits that any given fraction has is certainly countable, which makes the number of digits in these fractions to be $\aleph_0$. Since there are only two choices for what to put for a given digit, there must be $2^{\aleph_0}$ real numbers between 0 and 1, expressed in binary. Since we can disregard the binary (because every number written in binary can be written equivalently in base 10), we obtain the following: the number of reals between 0 and 1 must be $2^{\aleph_0}$. But we have already proven that the number of reals from between 0 and 1 is equivalent to the number of reals in a continuum $2^{\aleph_0} = \aleph_1$.

So, we can state the following:

$$\aleph_0 < \text{cardinality}(\mathbb{R}) = \text{cardinality}((0, 1)) = \text{cardinality}(\mathcal{P}(\mathbb{N})) = 2^{\aleph_0}.$$

7 The Continuum Hypothesis

So far, we have described the notion of two kinds of infinities in sets. First are those that are associated with natural numbers, integers, and rational numbers. These are known as countable sets, denoted by $\aleph_0$. The second type is associated with real numbers, also known as uncountable sets, or continuums, denoted by $\aleph_1$. Could there perhaps exist something in between those two types of infinite sets, that is, an infinite set of real numbers that does not have a one-to-one correspondence with natural numbers or a one-to-one correspondence with real numbers?

The continuum hypothesis states that there is no such set of numbers that exists in the realm of cardinality between natural numbers ($\aleph_0$) and real numbers ($\aleph_1$). Cantor could not prove the continuum hypothesis, and the problem remained unsolved in the 20th century. It was considered important particularly by David Hilbert, who placed it first on his famous list of open problems to be tackled in the 20th century.

The trouble with the hypothesis began in 1931, when the Austrian logician Kurt Gödel discovered that any set of axioms that serve as a foundation for mathematics will always be incomplete. There will always be problems that the foundational axioms can't resolve and facts that we cannot prove. As it pertains to the continuum hypothesis, Gödel believed that it is precisely that: a problem that is independent and cannot be solved with the foundational axioms of mathematics and set theory. (There are a total of 10 set theory axioms, known as ZFC, for Zermelo-Fraenkel axioms with the axiom of choice.)

In 1940, Gödel showed that you can't use the ZFC axioms to disprove the continuum hypothesis. And in 1963, through the efforts of American mathematician Paul Cohen, it was shown that the continuum hypothesis can't be proven either, that no axioms by mathematicians could be employed to solve this problem and the hypothesis was entirely independent of set theory and mathematics in general. For this, Cohen won a Fields Medal in 1966.

To do this, Paul Cohen used a technique called **forcing**. He began with a model that included a number line with $\aleph_0$ reals. Since you can "slice" this number line infinitely many times in infinitely many ways, one could slice the line on either side of some real number. This process could then be repeated with all other real numbers. And one could repeat this slicing procedure for each real number. But Cohen used a filter, so that this infinite slicing takes place within an interval that gets narrower and narrower. This begins with a slice that cuts the number line into two sections, and you pick one of them to work with, called Section A. The subsequent slice then comes from a point that is in Section A, and also creates a Section B. The third point for slicing can be found in the intersection of Sections A and B.

COHEN'S FILTER

Start with $\aleph_1$ numbers along the real number line.

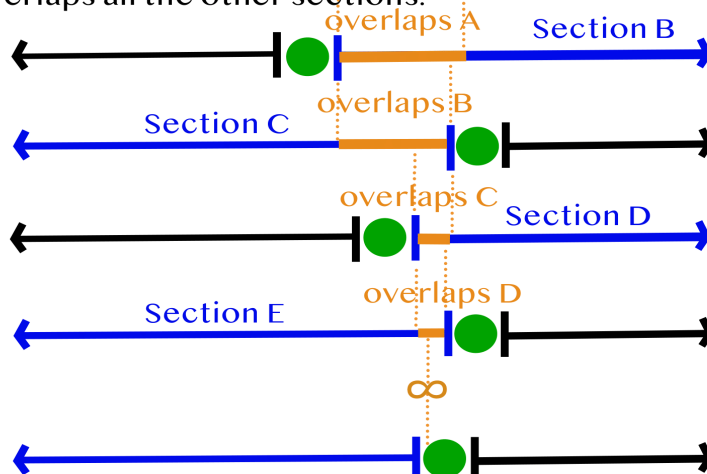
$-\infty \leftarrow \longrightarrow \infty$

Divide it into two sections that begin on either side of a

real number



Repeat this process for each real number. Choose each additional section making sure that it overlaps all the other sections.



The overlap of all the sections will be vanishingly small — a set consisting of a real number that can't possibly be one of the reals we started with. It's a new real number.

As the overlap of all of these sections continues, the remaining set gets smaller and smaller, until the overlap is nothing more than a single real number. But we also know that each real number where we performed the slicing was excluded from the sections that we produced. Therefore, the real number that was obtained at the end was not a real number that this slicing started with: It must be a new real number, which allows us to expand the set of with cardinality of $\aleph_1$ to include any number of reals, like $\aleph_2$ or $\aleph_{1.78}$.

But this seems to raise more questions than it answers. It produces the strange effect that any real number can be invented by any arbitrary forcing procedure, which seems undoubtedly confusing. Does an entity exist because it was invented by forcing?

8 Modern Discoveries and the Battle with the Continuum Hypothesis

These issues with forcing were resolved by creating another set of axioms for it. One important rule is called Martin's maximum, which states that any object that is created by forcing indeed

exists as a mathematical object if it satisfies certain conditions that show that it behaves consistently within the set of reals. Interestingly enough, despite the flexibility that this allows, it still seems to suggest that the set of reals moves up only by 1: to $\aleph_2$ and nothing greater.

Later in the 1990s, a mathematician named William Hugh Woodin introduced an axiom called (*) [star] that also tries to show that the continuum has a cardinality of $\aleph_2$. (*) involves a model universe of sets that satisfies the nine ZFC axioms plus the axiom of determinacy. A particularly important part of (*) is the statement that for all sets of $\aleph_1$ reals, there exist reals not in those sets, which is a clear negation of the continuum hypothesis.

The problem is that the foundational axioms with which Martin's maximum and (*) were constructed were the axioms of choice and determinacy, respectively, which contradict each other.

However, a new 2018 work by mathematician David Asperó shows that any extension of Martin's maximum, denoted Martin's maximum++, implies the correctness of (*). In order to do this, he used a procedure called L-forcing, and breaking it up into a sequence of recursive forcing procedures that satisfied the conditions of Martin's maximum.

Thus, with the newfound connection between Martin's maximum++ and (*), the case against the continuum hypothesis continues to build, as does the case for a continuum the size of $\aleph_2$. A final wrinkle: When he created (*), Woodin also stated some of its stronger counterparts, called (*)+ and (*)++, which apply to the power set of the set of reals. It's known that, in various models, (*)+ contradicts Martin's maximum. In a new proof, Woodin demonstrated that (*)+ and (*)++ are equivalent, meaning that (*)++ contradicts Martin's maximum as well. And because Martin's maximum seems to contradict (*)+ and (*)++, it raises doubts as to whether these theorems actually contradict the continuum hypothesis in the first place. Even Woodin himself admitted that he has given up hope on (*), and has rediscovered belief in the hypothesis. The battle for the truth about the continuum hypothesis continues to rage on.

9 Conclusion

This article examined the foundations of infinite set theory, including the sets of natural numbers, integers, rational numbers, and real numbers. This includes a comparison of the cardinalities of these infinite sets and the battle over the continuum hypothesis that continues to this day. The complexity of this field lies perhaps in its unconventional and abstract way of thinking. In that sense, this branch of mathematics shows us just how powerful math can be and what it can do: to understand, elucidate, and manipulate the power of infinity.

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